

Micro-grid For Hybrid Inverter+ On-grid Inverter System



Introduction

There is a large number of traditional on-grid inverters on the market. Due to islanding effect, on-grid inverters are unable to work during off-grid. This characteristic makes users lose the on-grid inverter PV energy when off-grid.

Micro-grid is the function that making hybrid inverter simulates the grid to activate on-grid inverter during off-grid. By connecting on-grid inverter to hybrid inverter's EPS port, hybrid inverter can use PV or battery energy to activate on-grid inverter during off-grid.

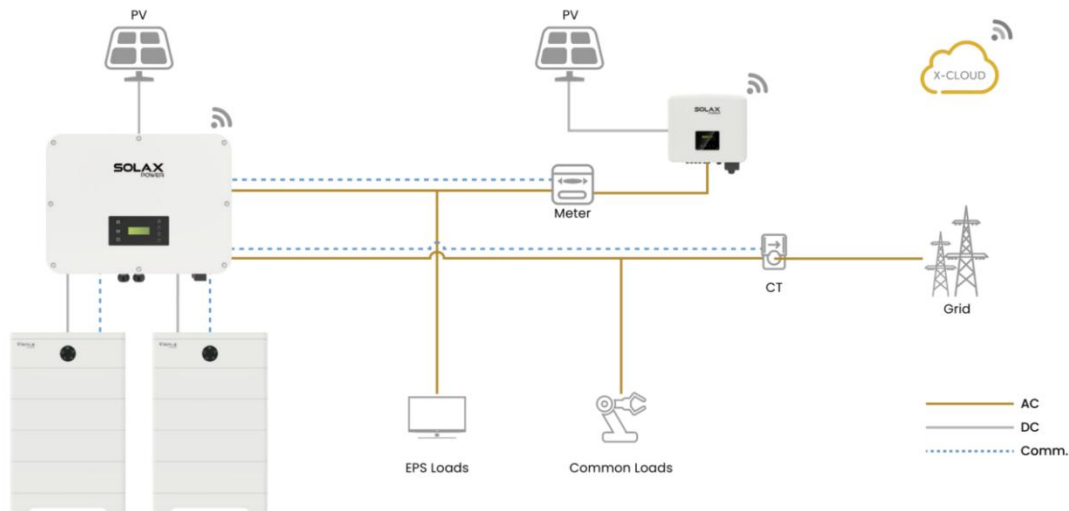


Diagram 1

Advantages

Cost-saving

Wasted on-grid inverter PV energy been saved, lower electricity bill

Eco-friendly

No extra devices needed

Simple and worry-free

No complicated configuration required, enable and forget

Configuration

1. Strictly follow the wiring diagram on Diagram 1.
2. Enable "Settings → Advanced Settings → Micro-grid".
3. Set "Settings → Advanced Settings → Micro-grid → PV INV Max Power"

PV INV MAX Power should be set to **on-grid inverter maximum output power. The value must be less than the rated power of the hybrid inverter. When the hybrid inverter begins to limit the current, causing the power at the EPS port to be lower than the set value, a 53Hz frequency signal will be sent to stop the operation of the PV inverter.*

Compatibility

Hardware version:

Hybrid inverter	X1-Hybrid/Fit G4	X1/3-IES	X1-Vast	X3-HYB-G4 Pro	X3-ULTRA	X1-SPT
On-grid inverter ^①	Single-phase	Single/ Single &Three-phase	Single-phase	Single &Three-Phase	Single &Three-phase	Single split-phase
Battery ^②	T-BAT-SYS-HV-5.8 T-BAT-SYS-HV-3.0 T-BAT-SYS-HV-R2. 5/3.6 T-BAT-SYS-HV-S2. 5/3.6	T-BAT-SYS-HV-S5 0E	T-BAT-SYS-HV-5.8 T-BAT-SYS-HV-3.0 T-BAT-SYS-HV-S2. 5/3.6	T-BAT-SYS-HV-5.8 T-BAT-SYS-HV-3.0 T-BAT-SYS-HV-S2. 5/3.6 T-BAT-SYS-HV-S5 0E	T-BAT-SYS-HV-5.8 T-BAT-SYS-HV-3.0 T-BAT-SYS-HV-R2. 5/3.6 T-BAT-SYS-HV-S2. 5/3.6 T-BAT-SYS-HV-S5. 1	T-BAT-SYS-HV-5.8 T-BAT-SYS-HV-S3. 6 (Only for X1-SPT 6.0/7.0)

Note:

- ① On-grid inverters can be any brand which supports “**power response to over frequency**”
- ② Compatible batteries may be adjusted, please contact service support for the latest battery compatibility file.

Software version:

Hybrid inverter	X1-Hybrid/ Fit G4	X1-IES	X3-IES	X1-Vast	X3-HYB- G4 Pro	X3-ULTRA	X1-SPT
Software version	ARM≥1.43 DSP≥1.46	ARM≥5.07 DSP≥10.01	ARM≥8.03 DSP≥11.00	ARM≥4.15 DSP≥6.12	ARM≥5.11 DSP≥5.12	ARM≥14.08 DSP≥13.02	ARM≥1.11 DSP≥1.11

Note:

1. X1-Fit G4 cannot be used in pure off-grid environment
2. The maximum output power of the On-grid inverter must meet the following two points **at the same time**:
 - 1) On-grid inverter output power ≤ Max. hybrid inverter EPS output power
 - 2) On-grid inverter output power ≤ Max. battery charging power
3. EPS Max. single-phase output power of the Hybrid inverter:

X3-IES	EPS Max. single-phase output power
5K	2.5kW
6K	3kW

8K	4kW
10K	5.5kW
12K	5.5kW
15K	5.5kW

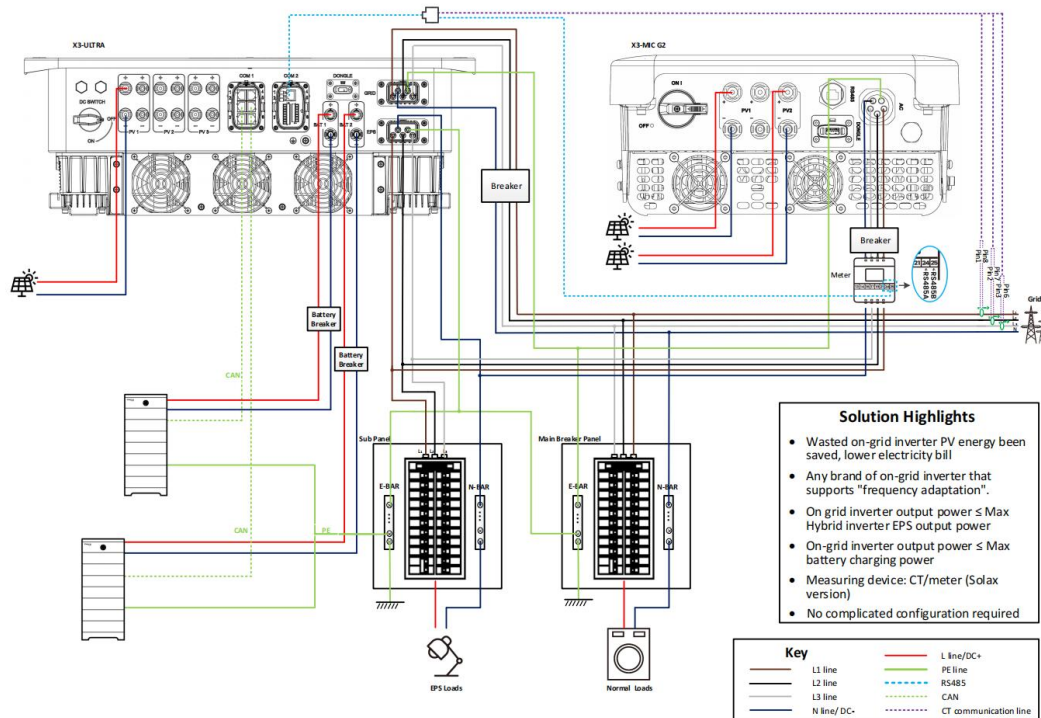
X3-HYB G4	EPS Max. single-phase output power
4K	2kW
5K	2.5kW
6K	3kW
8K	4kW
10K	5kW
12K	5.75kW
15K	5.75kW

X3-ULTRA	EPS Max. single-phase output power
15K	6.6kW
20K	6.6kW
25K	10kW
30K	10kW

Case study:

The following is an example of a micro-grid system:

Ultra Micro Grid Solution



Inverter Model & Quantity	Inverter Rated Power	Photovoltaic Panel	Battery
X3-Ultra*1	30kW	30kWp	/
T-BAT-SYS-HV-S3.6*13	/	/	47.8kWh
X3-Smart G2	10kW	12kWp	/

X3-Ultra in this system is paired with the following batteries system: the batteries are in a tower connected to two battery ports, charging capacity of the batteries is 39.9kW(13*51.2V*30A*2). Compared to X3-Ultra 30K EPS rated power, 30K is the batteries's Max. charging power (whichever is the lesser).

Therefore, this system satisfies $P_{on-grid\ INV} < P_{ultra\ max.\ Bat\ charging}$ and $P_{on-grid\ INV} =$

$P_{ultra\ EPS\ Max.\ single-phase\ output\ power}$. This is a successful micro-grid system.

4. Active response to frequency deviation with micro-grid function

In order to improve the stability of the micro-grid function, the logic of micro-grid frequency rise has been optimized[®], as follows:

- 1) When the micro-grid function is not enabled, the default frequency of the output voltage in off-grid mode is 50Hz/60Hz;
- 2) When the micro-grid mode is enabled and the battery does not limit the charging current (Taking 50Hz as an example, see Diagram 2):
 - a. When the SOC is less than 80%, the frequency of the output voltage in

off-grid mode is 50Hz;

- b. When the SOC is greater than 80%, then a frequency boost process is initiated;
- c. When the SOC exceeds 90%, the frequency boost process is accelerated at a higher rate; (Increasing the output voltage frequency to trigger the on-grid inverter's over-frequency load shedding logic)
(When the off-grid frequency is 60Hz, the corresponding node on Diagram 2 is 61Hz/63Hz)

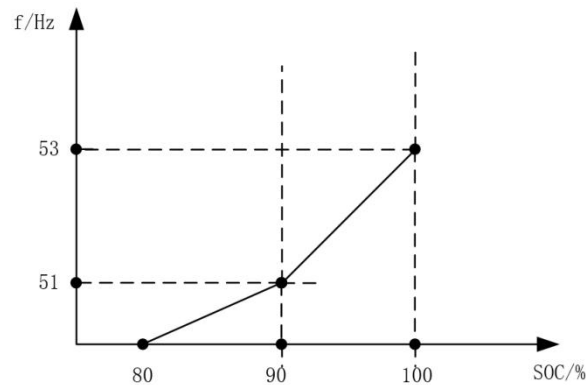


Diagram 2: The relationship between the battery's SOC and the frequency boost

- 3) When the micro-grid mode is enabled, if the maximum charging power limit of the battery is still greater than the maximum output power of the externally connected on-grid inverter, the output voltage in off-grid mode is 50Hz/60Hz;
- 4) When the micro-grid mode is enabled, if the maximum charging power limit of the battery is less than the maximum output power of the externally connected on-grid inverter, the frequency of the output voltage in off-grid mode is 53Hz/63Hz, which will limit the operating power of the on-grid inverter^④;

Note:

③ This micro-grid logic has been implemented on software version X1-Hybrid/Fit G4/X1-IES/X3-IES/X3-ULTRA mentioned above.

④ In a micro-grid system, the hybrid inverter is unable to ascertain the actual output power of the on-grid inverter. If the maximum output power of the on-grid inverter is close to the maximum charging power of the battery, and when the battery's charging current is limited due to factors such as temperature, the micro-grid function will not be able to operate normally in order to ensure system safety.
